



# Kinetic-hybrid Simulations of Counterpropagating Ion Cyclotron Waves and Proton Scattering in the Near-Sun Solar Wind

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Received 2025 January 16; revised 2025 February 16; accepted 2025 March 2; published 2025 April 11

## Abstract

Ion cyclotron waves (ICWs) are prevalent in the near-Sun solar wind and play a significant role in the nonadiabatic heating of plasma. Recent observations from the Parker Solar Probe have revealed the simultaneous presence of antisunward and sunward ICWs in the vicinity of the Alfvén surface. However, single-satellite observations cannot effectively trace the generation and evolution of these observed waves. To address this limitation, we employ kinetic-hybrid simulations to replicate the generation and evolution of counterpropagating ICWs under typical plasma conditions in the near-Sun solar wind. Following the linear growth phase, the simulated waves exhibit amplitude and polarization characteristics that closely match the observations. Additionally, our simulation illustrates proton scattering induced by the counterpropagating waves. These results underscore the significance of locally generated ICWs in influencing solar wind ion dynamics.

*Unified Astronomy Thesaurus concepts:* [Solar wind \(1534\)](#); [Plasma physics \(2089\)](#); [Space plasmas \(1544\)](#)

## 1. Introduction

Two fundamental problems in the heliophysics community remain unresolved (e.g., E. Marsch 2006; S. R. Cranmer et al. 2017): how the fast solar wind is accelerated and how the solar coronal plasma is heated. Although both phenomena are believed to be driven by various collisionless particle energization processes (e.g., J. V. Hollweg & J. M. Turner 1978; J. V. Hollweg & P. A. Isenberg 2002; Y. Voitenko & M. Goossens 2004; B. D. G. Chandran et al. 2010, 2011), such as wave–particle interactions and magnetic reconnection, direct observational evidence supporting the proposed models has been lacking prior to the in situ measurements made possible by the Parker Solar Probe (PSP; N. Fox et al. 2016), launched on 2018 August 12. Among the proposed wave–particle interaction models, ion cyclotron waves (ICWs) have been the most frequently discussed wave mode for understanding ion energization (J. V. Hollweg & P. A. Isenberg 2002). By incorporating the cyclotron resonant process associated with ICWs into solar wind models (e.g., J. V. Hollweg & J. M. Turner 1978; E. Marsch et al. 1982; P. A. Isenberg & J. V. Hollweg 1983; Y. Q. Hu et al. 1997; C. Y. Tu & E. Marsch 1997; Y. Q. Hu & S. R. Habbal 1999; X. Li et al. 1999; B. Li et al. 2004; S. R. Cranmer 2000; P. A. Isenberg et al. 2001; J. V. Hollweg & P. A. Isenberg 2002), these models can provide reasonable explanations for the radial distribution of the bulk flow and temperature of the ion population (e.g., E. Marsch et al. 1982; J. L. Kohl et al. 1998), as supported by previous in situ (e.g., Helios) and remote observations (e.g., SOHO) prior to the PSP era. Recent observations from the PSP have revealed the widespread presence of ICWs in the near-Sun

solar wind (T. A. Bowen et al. 2020a, 2020b; J. Huang et al. 2020; J. L. Verniero et al. 2020, 2022; K. G. Klein et al. 2021; C. Shi et al. 2021; M. D. McManus et al. 2024), with the occurrence of these waves reaching approximately 30% near the Sun (W. Liu et al. 2023). The observations also suggest that ICWs significantly contribute to the nonadiabatic heating of the near-Sun solar wind plasma (T. A. Bowen et al. 2022, 2024; N. Shankarappa et al. 2024). Consequently, research on the generation and evolution of ICWs and their role in particle dynamics within the plasma environment close to the Sun has become a focal topic in contemporary studies.

A common physical process for the generation of ICWs is ion instability (e.g., S. P. Gary 1993; H. Sun et al. 2019), due to unstable ion velocity distributions function (VDF). In the case of quasi-monochromatic ICWs observed in the near-Sun solar wind (e.g., T. A. Bowen et al. 2020a, 2020b; J. L. Verniero et al. 2020, 2022; W. Liu et al. 2023), these waves are often accompanied by a proton beam component with a considerable number density (J. L. Verniero et al. 2020, 2022). The beam protons flow at a higher velocity than the background core protons, creating a relative drift between the two proton components that can provide free energy for ICW generation (e.g., M. D. Montgomery et al. 1976; W. Daughton & S. P. Gary 1998; W. Liu et al. 2021; J. L. Verniero et al. 2020; L. Ofman et al. 2022). However, this proton beam instability primarily produces ICWs that propagate in the same direction as the antisunward proton beam (e.g., T. A. Bowen et al. 2020b; W. Liu et al. 2023). Correspondingly, the generated ICWs can only affect the behavior of sunward ions in the solar wind frame through cyclotron resonant wave–particle interactions (J. V. Hollweg & P. A. Isenberg 2002).

Recently, C. Shi et al. (2024) reported a novel event involving ICWs where both antisunward and sunward waves coexist in the near-Sun solar wind. These counterpropagating ICWs are believed to be generated by proton cyclotron



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instabilities stemming from temperature anisotropy—specifically, the temperature parallel to the magnetic field is smaller than that in the perpendicular direction—within the proton beam and core populations (C. Shi et al. 2024). The significance of these counterpropagating ICWs lies in their ability to resonantly scatter both antisunward and sunward ions in the near-Sun solar wind. However, cyclotron resonant interactions with antisunward ions have received limited attention in previous solar wind models.

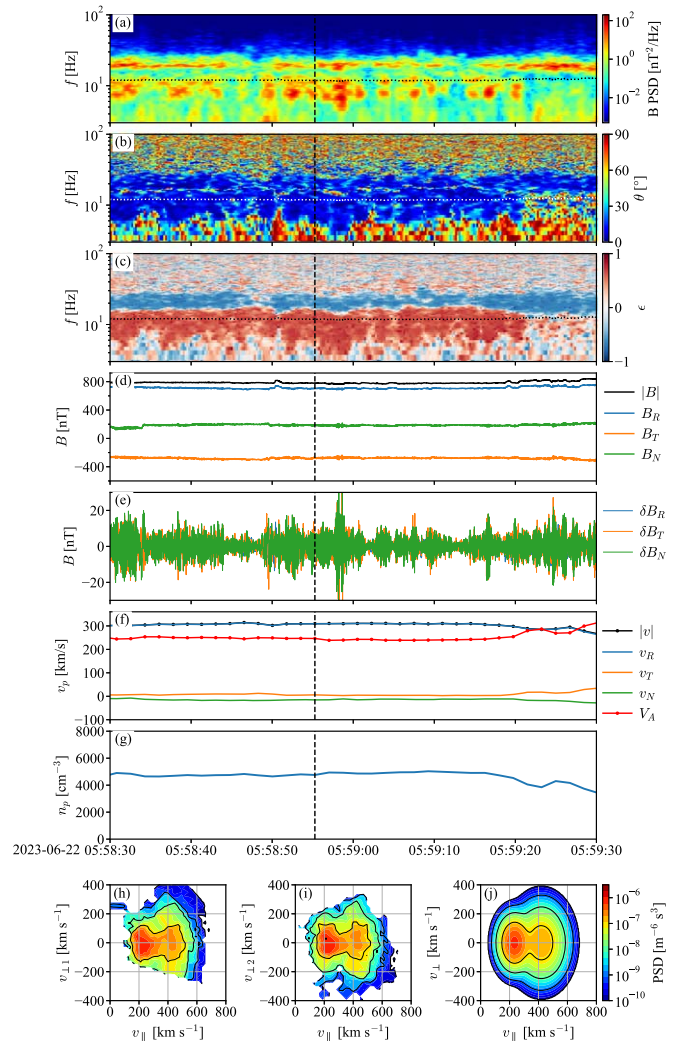
The C. Shi et al. (2024) study is mainly based on linear growth rate calculation using a single-point observation; therefore, their analysis does not address the wave evolution and saturation processes. Furthermore, the detailed wave–particle interaction process due to the excitation of the observed counterpropagating ICWs could not be investigated. On the other hand, quasilinear models cannot describe potential nonlinear wave–particle interactions with large-amplitude waves or wave–wave interactions (e.g., L. Ofman & A. F. Viñas 2007; L. Ofman et al. 2014). To address these limitations in the observational study, this paper employs kinetic-hybrid particle-in-cell (PIC) simulations to investigate the generation and evolution of counterpropagating ICWs based on the event observed by C. Shi et al. (2024). Such kinetic-hybrid PIC simulations have been extensively used to investigate the wave–particle interactions associated with ICWs and other wave dynamics (e.g., Y. Omura et al. 1985; S. P. Gary & S. Saito 2003; H. Xie et al. 2004; S. P. Gary et al. 2005; X. Li & S. R. Habbal 2005; L. Ofman & A. F. Viñas 2007; L. Ofman et al. 2014; N. Omididi et al. 2014; K. Min et al. 2015; L. Ofman 2019; S. A. Markovskii & B. J. Vasquez 2023, 2024). While previous studies have primarily investigated ICWs driven by proton beam instabilities (L. Ofman et al. 2022, 2023; L. Pezzini et al. 2024), simulation research of ICWs due to proton cyclotron instability based on plasma parameters in the solar corona and near-Sun solar wind remains limited. Therefore, our simulation aims to investigate the excitation and evolution of counterpropagating ICWs due to cyclotron instability and offer insights into the related proton energization process.

This paper is organized as follows: First, we briefly present the typical wave and plasma parameters associated with the ICW event reported by C. Shi et al. (2024) in Section 2. Next, Section 3 describes the settings for our kinetic-hybrid PIC simulations. Section 4 shows the results concerning the generation and evolution of counterpropagating ICWs, as well as the proton scattering induced by these waves in our simulation. Finally, Section 5 presents the conclusions and discussions from this study.

## 2. Wave and Plasma Parameters from Observations

The ICW event of interest was reported by C. Shi et al. (2024). This event is located at approximately  $13.3 R_{\odot}$ , near the Alfvén surface. Figure 1 provides an overview of the wave and plasma environment, based on measurements from the flux-gate magnetometer (MAG; S. Bale et al. 2016) and the Solar Wind Electrons Alphas and Protons (SWEAP) instrument (J. C. Kasper et al. 2016) onboard the PSP.

Figures 1(a)–(c) display the wave properties obtained through polarization analysis of the fast Fourier transform (FFT) data of time-series magnetic fields (J. D. Means 1972). From Figure 1(a), we observe that the power spectral densities of the magnetic field ( $P_B$ ) are enhanced in two distinct frequency ranges: 6–15 and 15–25 Hz. The waves in both



**Figure 1.** Overview of the observed ICW event. (a) Time–frequency power spectrogram of magnetic field waves; (b) wave normal angle ( $\theta$ ); (c) ellipticity ( $\epsilon$ ); (d) observed magnetic field vector in RTN coordinates and total magnetic field strength  $|B|$ ; (e) waveform of perturbed magnetic fields ( $>1$  Hz); (f) proton velocity in RTN coordinates and Alfvén velocity ( $V_A$ ); (g) proton density  $n_p$ ; (h)–(i) observed proton VDFs at two different cut planes; (j) the fitted proton VDF as given by C. Shi et al. (2024). The horizontal dotted line in panels (a)–(c) indicates proton gyrofrequency, while the vertical dashed line indicates the time at which the proton VDFs are observed.

frequency bands exhibit a small normal angle ( $\theta \sim 0^\circ$ ; Figure 1(b)), yet they demonstrate different polarization characteristics (Figure 1(c)). Specifically, the upper-band waves are left-handed polarized ( $\epsilon < 0$ ), while the lower-band waves are right-handed polarized ( $\epsilon > 0$ ).

Figures 1(d), (f), and (g) display the plasma environment associated with the two-band waves. Both the magnetic field (Figure 1(d)) and the proton bulk flow (Figure 1(f)) are primarily directed radially outward from the Sun. Since the local Alfvén speed  $V_A$  is approximately equal to the bulk flow speed  $|v_p|$ , the counterpropagating ICWs are distributed near the Alfvén surface. The proton number density remains nearly constant (Figure 1(g)), suggesting that the observed waves exist within a homogeneous background environment. Additionally, the magnetic field fluctuations, shown in Figure 1(e), are obtained by applying a high-pass filter (above 1 Hz), revealing wave amplitudes of approximately 5 to 20 nT.

Figures 1(h)–(j) further illustrate the observed and fitted proton VDFs at a typical time (05:58:55.27 UT) in field-aligned coordinates. The three axes are defined as  $\hat{e}_{\parallel} = \mathbf{B}/|\mathbf{B}|$ ,  $\hat{e}_{\perp 1} = \hat{e}_{\parallel} \times \hat{e}_x$  (where  $\hat{e}_x$  represents the unit vector along the rotation axis of the SPAN-I instrument), and  $\hat{e}_{\perp 2} = \hat{e}_{\parallel} \times \hat{e}_{\perp 1}$ . A primary feature of the proton VDFs measured by SWEAP/SPAN-I (R. Livi et al. 2022) is the presence of two distinct proton populations with different drift speeds in the  $\hat{e}_{\parallel}$  direction, as shown in Figures 1(h) and (i). Moreover, the proton VDFs in the  $\hat{e}_{\parallel} - \hat{e}_{\perp 1}$  and  $\hat{e}_{\parallel} - \hat{e}_{\perp 2}$  planes are similar, implying that the protons generally exhibit a gyrotropic distribution.

By employing a two-proton-component model and assuming that the components follow a drifting bi-Maxwellian distribution, C. Shi et al. (2024) obtain a fitted proton VDF consisting of a slow-streaming proton core component and a fast-streaming beam component, as depicted in Figure 1(j). The fitted plasma parameters are as follows: for the proton core component, the number density  $n_{pc} = 2530 \text{ cm}^{-3}$ , the drift velocity  $v_{pc} = 234 \text{ km s}^{-1}$ , the parallel temperature  $T_{pc\parallel} = 20 \text{ eV}$ , and the perpendicular temperature  $T_{pc\perp} = 52 \text{ eV}$ ; for the proton beam component,  $n_{pb} = 2169 \text{ cm}^{-3}$ ,  $v_{pb} = 420 \text{ km s}^{-1}$ ,  $T_{pb\parallel} = 48 \text{ eV}$ , and  $T_{pb\perp} = 107 \text{ eV}$ .

This fitted VDF effectively captures the observed anisotropic characteristics of both proton components, and previous studies have shown both components have sufficient temperature anisotropy to drive ICWs based on linear theory analysis. Therefore, the observed two-band wave nature has been proposed as counterpropagating ICWs in the plasma frame (C. Shi et al. 2024). In this context, the sunward ICWs in the plasma frame are convected by the super-Alfvénic solar wind, leading to a reversal of the wave propagation direction (antisunward propagation) and a change in wave polarization (right-hand polarization) in the spacecraft frame (L. K. Jian et al. 2009; J. L. Verniero et al. 2020; J. Zhao et al. 2020; C. Shi et al. 2024).

### 3. Simulation Settings

To simulate the observed ICWs and relevant ion dynamics, we use the open-source PIC simulation code, WarpX (J.-L. Vay et al. 2018; L. Fedeli et al. 2022). This code supports a kinetic-hybrid model in which the kinetic effects of ions are considered while electrons are treated as a fluid governed by a simplified Ohm’s law (R. E. Groenewald et al. 2023). This model is particularly useful for studying ion-scale processes such as ICWs, as they are only weakly affected by electron kinetic effects. A significant advantage of the hybrid approach is that it greatly reduces computational costs by allowing larger time steps and grid sizes.

Given that the observed ICWs propagate nearly parallel to the background magnetic field, we choose a one-dimensional simulation domain along the  $z$ -axis with a uniform background magnetic field  $\mathbf{B} = B_0 \hat{e}_z$  and periodic boundary conditions. This choice is also supported by the results of the linear instability analysis under the plasma parameters in our simulation, which demonstrate that the strongest instability occurs in the parallel direction. In our setup, the time step is set to  $\Delta t = 0.05 \Omega_p^{-1}$ , ensuring sufficient resolution of the ion cyclotron motion. The spatial resolution is defined by a grid size of  $\Delta z = 0.3125 d_p$ , where  $d_p$  is the proton inertial length given by  $d_p \equiv c/\omega_{pp}$ , with  $c$  representing the speed of light and  $\omega_{pp}$  the proton plasma frequency. The simulation domain

consists of 512 grid cells, resulting in a total length of  $160 d_p$ , which is more than 10 times the ICW wavelength. To minimize numerical noise, we initialize each cell with  $2^{18} = 262144$  particles per species. Additionally, a finite resistivity of  $\eta = 13 \Omega m$  is incorporated into the simulations to maintain numerical stability. This approach is commonly used in hybrid simulations to eliminate nonphysical heating caused by unresolved high-frequency waves, such as whistler waves (e.g., S. P. Gary et al. 2003). While the resistivity may induce minor damping of ICWs, it does not significantly affect the wave–particle interactions central to this study.

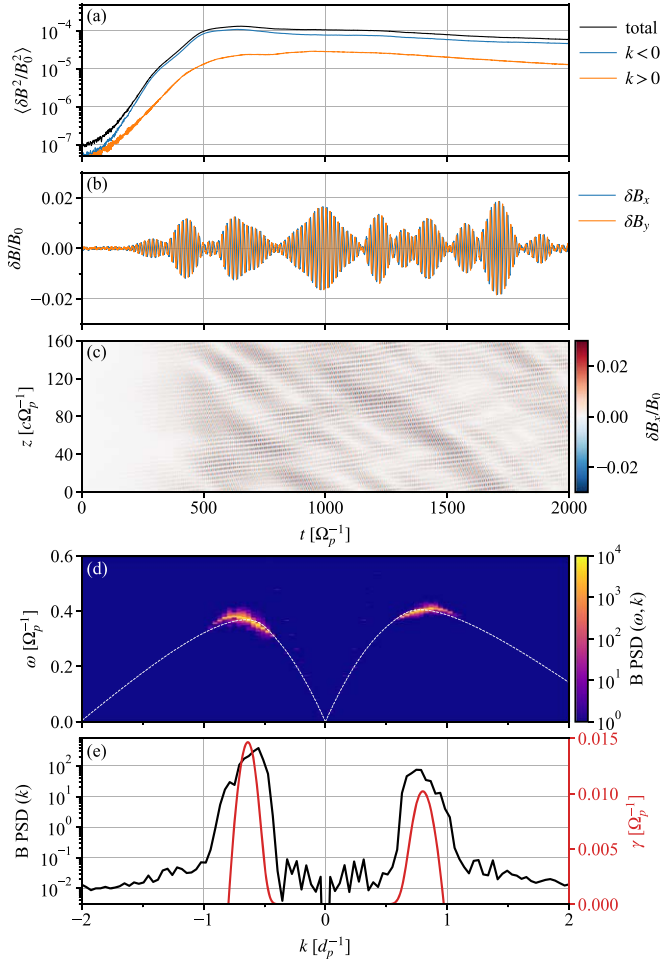
Based on the typical wave and plasma environment illustrated in Figure 1, we consider a three-component plasma consisting of a proton core, a proton beam, and electrons, with a homogeneous magnetic field ( $B_0 = 780 \text{ nT}$ ) and densities of  $n_{pc} = 2530 \text{ cm}^{-3}$  and  $n_{pb} = 2169 \text{ cm}^{-3}$ . We also adopt the solar wind frame with a solar wind speed of  $V_{SW} = 319.9 \text{ km s}^{-1}$ , resulting in drift speeds for the two proton components of  $v_{pc} = -85.9 \text{ km s}^{-1}$  and  $v_{pb} = 100.1 \text{ km s}^{-1}$ . The drift speed of the electron population is assumed to be zero to maintain an overall current-free system.

Due to the considerable amplitude of the observed ICWs, they are expected to strongly scatter protons and reduce the temperature anisotropy. In our simulation, to better align the simulated wave amplitude with observations, the perpendicular temperatures are assumed to be 40% larger than the fitted values from observations, while the parallel temperatures remain the same as the fitted values. The parallel beta  $\beta_{\parallel}$  and temperature anisotropy  $A = T_{\perp}/T_{\parallel} - 1$  of the two proton components are as follows: for core protons,  $\beta_{pc\parallel} \simeq 0.033$  and  $A_{pc} \simeq 2.64$ ; and for beam protons,  $\beta_{pb\parallel} \simeq 0.069$  and  $A_{pb} \simeq 2.12$ . Both temperature anisotropies exceed the threshold anisotropy required to drive ion cyclotron instability (e.g., S. P. Gary et al. 1994), indicating that ICWs can be produced by both proton components. We have also conducted simulations using lower temperature anisotropy (not shown) and observed the excitation of counterpropagating ICWs via the proton cyclotron instability, albeit with weaker wave amplitudes.

## 4. Simulation Results

### 4.1. The Generation and Evolution of the Counterpropagating ICWs

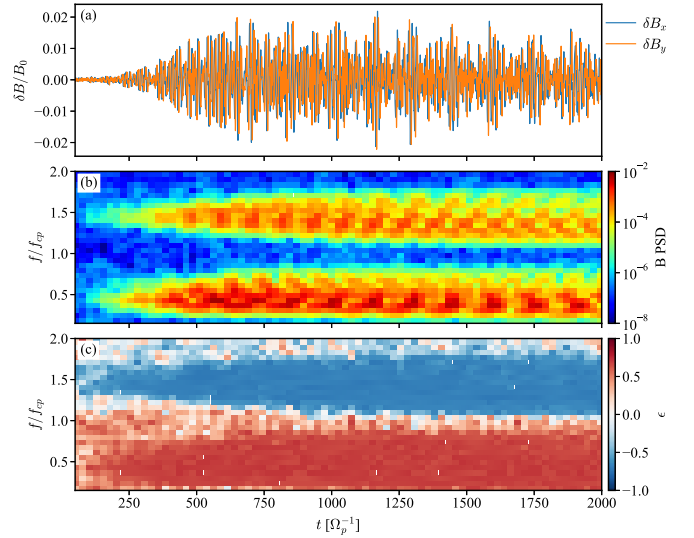
Figure 2 illustrates the simulation results about the generation and evolution of the waves in the solar wind frame. The evolution of wave magnetic field intensity is displayed in Figure 2(a). We separate the waves into antisunward ( $k > 0$ , red) and sunward ( $k < 0$ , blue) components using the method proposed by T. Terasawa et al. (1986). From Figure 2(a), we observe that the sunward waves exhibit faster growth and stronger intensity compared to the antisunward waves. Both types of waves undergo two distinct evolutionary stages. Before  $t \sim 500 \Omega_p^{-1}$ , the waves grow linearly, reaching a peak intensity of  $\delta B^2 = 1.34 \times 10^{-4} B_0^2$  at the end of this linear phase. The corresponding wave amplitude is approximately  $0.0116 B_0 \simeq 9 \text{ nT}$ , consistent with the observational values shown in Figure 1(e). After  $t \sim 500 \Omega_p^{-1}$ , the waves propagating to both directions reach saturation, with minor wave energy variation. Figure 2(b) shows the waveform evolution recorded at  $z = 0$  in the simulation. It reveals that the excited waves behave as wave packets with various envelope forms, similar to



**Figure 2.** Ion cyclotron waves generated in PIC simulations. (a) Temporal evolution of sunward ( $k < 0$ ), antisunward ( $k > 0$ ), and total wave intensity; (b) the magnetic field waveform recorded at  $z = 0$ ; (c) the spatiotemporal map of the  $B_x$  wave component; (d) the two-dimensional power spectral density (PSD) in the  $\omega$ - $k$  plane, with the white dashed curve showing the linear dispersion relation of ICWs; (e) the wavenumber power spectrum density  $P_B(k)$  (black) and the linear growth rate.

the observed waves presented in Figure 1(e). Figure 2(c) further illustrates the spatiotemporal evolution of the wave fields in the  $z$ - $t$  plane, displaying interlaced wavefronts during the saturation stage, which are caused by the generation of counterpropagating waves.

We explain the generation mechanism of the simulated waves in Figures 2(d) and (e). By applying a two-dimensional FFT to the wave field data  $\delta \mathbf{B}(t, z)$ , we obtain the power spectral density of the magnetic field,  $P_B$ , in the  $\omega$ - $k$  space, as shown in Figure 2(d). This representation clearly indicates the presence of counterpropagating waves, with  $\omega/\Omega_p \sim 0.4$  and  $kd_p \sim \pm 0.7$ . To elucidate the generation mechanism for these waves, we solve the linear kinetic dispersion relation of the wave using the BO code (H. Xie & Y. Xiao 2016; H. Xie 2019). The theoretical instability analysis successfully predicts the existence of unstable ICWs propagating both antiparallel and parallel to the background magnetic field. The dispersion relations of these unstable ICWs are overlaid in Figure 2(d), showing that the simulated waves are consistent with the theoretical predictions for both sunward and antisunward ICWs. We integrate  $P_B(\omega, k)$  over the frequency  $\omega$  to obtain  $P_B(k)$  in  $k$ -space, as shown in Figure 2(e). The wavenumber  $k$



**Figure 3.** ICWs observed by the virtual spacecraft in the simulation. (a) Waveform of the magnetic field; (b) time-frequency spectrogram; (c) ellipticity of waves.

of the two peaks of  $P_B(k)$  roughly corresponds to the maximum linear growth rates of antisunward and sunward ICWs predicted by theory. Moreover, linear theory also predicts that the growth rate of the sunward waves is larger than that of the antisunward waves, consistent with results from our simulations.

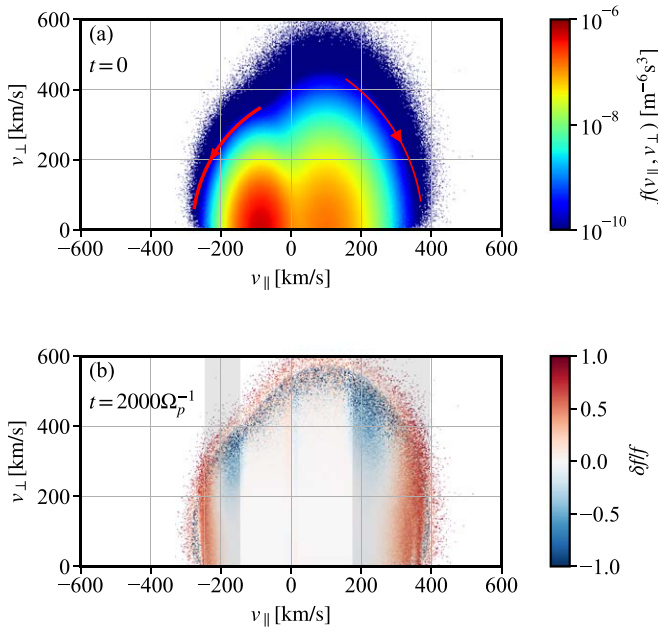
To compare between our simulation results and observations, we conduct a polarization analysis on the simulated data observed by a virtual spacecraft. The virtual spacecraft records the wave magnetic field and its location is given by  $z_s = -tV_{SW} + z_0$ , where  $z_0$  is the virtual spacecraft's initial position. This method allows the detection of the magnetic field fluctuations in the spacecraft frame, similar to in situ spacecraft measurements. For simplicity, we set  $z_0$  at the uppermost boundary of the simulation domain and allow the virtual spacecraft to traverse the simulation system multiple times across the periodic boundary.

Based on the simulated magnetic fluctuations in the virtual spacecraft frame, shown in Figure 3(a), we generate a time-frequency spectrogram, shown in Figure 3(b). This spectrogram reveals the presence of two-band waves, with frequency ranges approximately between  $0.2$ – $0.9f_{cp}$  and  $1.1$ – $1.8f_{cp}$ , corresponding well to observational results. Furthermore, the polarization of these two-band waves is presented in Figure 3(c). The upper-band waves exhibit left-hand polarization, while the lower-band waves exhibit right-hand polarization, consistent with observational findings.

The consistency between simulation results and observations suggests that the simulation successfully reproduces the observations of counterpropagating ICWs presented in Figure 1, thereby validating the generation mechanism proposed by C. Shi et al. (2024). Furthermore, it allows us to investigate the scattering of protons in the generation process of the ICWs.

#### 4.2. The Proton Scattering by the Counterpropagating ICWs

Due to cyclotron resonant interactions, protons can be scattered in different directions in the presence of counterpropagating ICWs. The signatures of resonant protons and their scattering paths are displayed in Figure 4.



**Figure 4.** (a) The initial proton VDF with red arrows showing the diffusion paths of beam (thick curve) and core proton populations (thin curve); (b) the normalized variation in the proton VDF between  $t = 2000\Omega_p^{-1}$  and  $t = 0$ , with the resonant velocity ranges shaded in gray.

Based on the cyclotron resonant condition, the parallel velocities of resonant protons are determined by  $v_{\parallel} = (\omega - \Omega_p)/k$ . To estimate  $v_{\parallel}$  related to the simulated waves, we consider the wave frequency  $\omega$  and wavenumber  $k$  for waves with  $P_B(\omega, k) > 10^2$  (see Figure 2(d)). The predicted  $v_{\parallel}$  values are 176–395  $\text{km s}^{-1}$  for sunward ICWs and –245 to –146  $\text{km s}^{-1}$  for antisunward ICWs. These predicted  $v_{\parallel}$  regions are highlighted as gray areas in Figure 4(b), which presents the difference between the two proton VDFs at  $t = 0$  and  $t = 2000\Omega_p^{-1}$ , i.e.,  $\delta f = f(t = 2000\Omega_p^{-1}) - f(t = 0)$ .

On the other hand, the direction of resonant proton flux is determined by the gradient of the phase space density along the diffusion path. The scattering path of resonant protons in the velocity space is described by  $(v_{\parallel} - \omega/k_{\parallel})^2 + v_{\perp}^2 = C$ , indicating energy conservation of resonant protons in the wave frame. The sketch of the scattering path of resonant protons is illustrated in Figure 4(a). For resonant protons associated with sunward ICWs, the scattering direction is represented by the thick arrow, resulting from the gradient of the phase space density in the proton beam population. Conversely, the direction of resonant proton flux associated with antisunward ICWs is shown as the thin arrow, due to the gradient of the phase space density of the proton core population.

The effects of the scattering are shown in Figure 4(b): the resonant protons move from blue regions (characterized by smaller  $|v_{\parallel}|$  and larger  $v_{\perp}$ ) into red regions (characterized by larger  $|v_{\parallel}|$  and smaller  $v_{\perp}$ ). The consequence of this proton scattering is a reduction in temperature anisotropy within the proton population, leading to the saturation of the simulated waves.

## 5. Discussions and Conclusions

Our simulation indicates that the timescale  $T_{\text{LIN}}$  in the linear growth stage is approximately  $500\Omega_p^{-1}$ . This timescale is

roughly consistent with that observed in the proton cyclotron instability simulation performed by S. P. Gary et al. (2003), which considered an instability driven by a stronger temperature anisotropy of the proton population in the solar wind near 1 au. Based on the solar wind speed in the near-Sun environment,  $V_{\text{SW}} \sim 320 \text{ km s}^{-1}$  (see Figure 1(f)), the spatial scale  $L_{\text{LIN}}$  associated with  $T_{\text{LIN}}$  is about  $2.1 \times 10^3 \text{ km}$ , significantly smaller than the characteristic scale, the solar radius ( $R_{\odot} = 7 \times 10^5 \text{ km}$ ). This suggests that the assumption of a homogeneous background is valid in the simulation setting, allowing us to reasonably neglect the radial expansion effect in the model (e.g., P. Hellinger et al. 2003), which induces adiabatic cooling of the proton population.

To summarize, we used hybrid-PIC simulations to investigate the generation and evolution of counterpropagating ICWs in the near-Sun solar wind. Utilizing typical plasma parameters obtained from observations, our simulation successfully reproduced the ICWs with wave properties (i.e., amplitude, frequency, and polarization) similar to those observed by the PSP. Our results confirm the generation mechanism of the counterpropagating ICWs proposed by C. Shi et al. (2024), wherein sunward and antisunward ICWs are driven by the temperature anisotropy of the proton beam and core components, respectively. Furthermore, this study investigated the detailed scattering process of resonant protons by counterpropagating ICWs. Protons near resonant velocities are scattered by ICWs, moving from higher to lower pitch angles along diffusion paths. This scattering reduces temperature anisotropy, leading to the saturation of ICWs. This process affects both beam and core protons due to the bidirectional propagation of the waves. These results should be useful to understand wave generation and particle scattering processes in the near-Sun plasma.

## Acknowledgments

This research is supported by the Strategic Priority Research Program of the Chinese Academy of Sciences (grant No. XDB0560000), NSFC grant 42404162, and Postdoctoral Fellowship Program of CPSF under grant No. GZC20241625. Z.J.S. thanks for the support from NSFC 42374196. This research used the open-source particle-in-cell code WarpX (<https://github.com/ECP-WarpX/WarpX>). Primary WarpX contributors are with LBNL, LLNL, CEA-LIDYL, SLAC, DESY, CERN, and TAE Technologies.

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